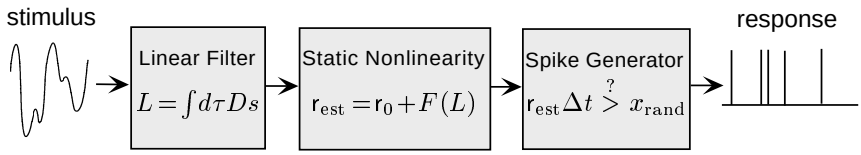

Figure 2.3 Simulating spiking responses to stimuli. The integral of the stimulus s times the optimal kernel D is first computed. The estimated firing rate is the background rate r_0 plus a nonlinear function of the output of the linear filter calculation. Finally, the estimated firing rate is used to drive a Poisson process that generates spikes.

not the estimate with the static nonlinearity $r_{est}(t) = r_0 + F(L(t))$. A theorem due to Bussgang (see appendix C) suggests that equation 2.6 will provide a reasonable kernel, even in the presence of a static nonlinearity, if the white noise stimulus used is Gaussian.

In some cases, the linear term of the Volterra series fails to predict the response even when static nonlinearities are included. Systematic improvements can be attempted by including more terms in the Volterra or Wiener series, but in practice it is quite difficult to go beyond the first few terms. The accuracy with which the first term, or first few terms, in a Volterra series can predict the responses of a neuron can sometimes be improved by replacing the parameter s in equation 2.7 with an appropriately chosen function of s, so that

$$L(t) = \int_0^\infty d\tau D(\tau) f(s(t-\tau)) \,. \tag{2.12}$$

A reasonable choice for this function is the response tuning curve. With this choice, the linear prediction is equal to the response tuning curve, $L = f(s)$, for static stimuli, provided that the integral of the kernel D is equal to 1. For time-dependent stimuli, we can think of equation 2.12 as a dynamic extension of the response tuning curve.

A model of the spike trains evoked by a stimulus can be constructed by using the firing-rate estimate of equation 2.8 to drive a Poisson spike generator (see chapter 1). Figure 2.3 shows the structure of such a model with a linear filter, a static nonlinearity, and a stochastic spike generator. In the figure, spikes are shown being generated by comparing the spiking probability $r(t)\Delta t$ to a random number, although the other methods discussed in chapter 1 could be used instead. Also, the linear filter acts directly on the stimulus s in figure 2.3, but it could act instead on some function $f(s)$, such as the response tuning curve.

2.3 Introduction to the Early Visual System

Before discussing how reverse-correlation methods are applied to visually responsive neurons, we review the basic anatomy and physiology of the

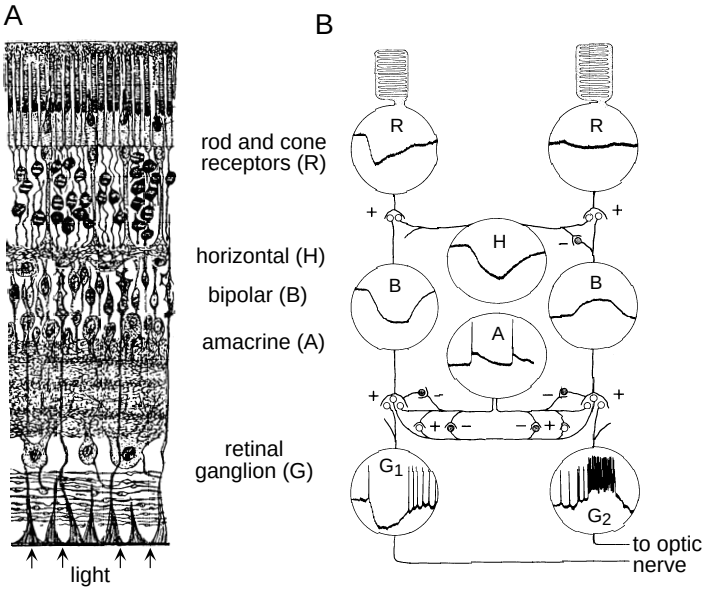

Figure 2.4 (A) An anatomical diagram of the circuitry of the retina of a dog. Cell types are identified at right. In the intact eye, counterintuitively, light enters through the side opposite from the photoreceptors. (B) Intracellular recordings from retinal neurons of the mud puppy responding to a flash of light lasting for 1 s. In the column of cells on the left side of the diagram, the resulting hyperpolarizations are about 4 mV in the rod and retinal ganglion cells, and 8 mV in the bipolar cell. Pluses and minuses represent excitatory and inhibitory synapses, respectively. (A adapted from Nicholls et al., 1992; drawing from Cajal, 1911. B data from Werblin and Dowling 1969; figure adapted from Dowling, 1992.)

early stages of the visual system. The conversion of a light stimulus into an electrical signal and ultimately an action potential sequence occurs in the retina. Figure 2.4A is an anatomical diagram showing the five principal cell types of the retina, and figure 2.4B is a rough circuit diagram. In the retina, light is first converted into an electrical signal by a phototransduction cascade within rod and cone photoreceptor cells. Figure 2.4B shows intracellular recordings made in neurons of the retina of a mud puppy (an amphibian). The stimulus used for these recordings was a flash of light falling primarily in the region of the photoreceptor at the left of figure 2.4B. The rod cells, especially the one on the left side of figure 2.4B, are hyperpolarized by the light flash. This electrical signal is passed along to bipolar and horizontal cells through synaptic connections. Note that in one of the bipolar cells, the signal has been inverted, leading to depolarization. These smoothly changing membrane potentials provide a graded representation of the light intensity during the flash. This form of coding is adequate for signaling within the retina, where distances are small. However, it is inadequate for the task of conveying information from the retina to the brain.

retinal ganglion cells

The output neurons of the retina are the retinal ganglion cells, whose axons form the optic nerve. As seen in figure 2.4B, the subthreshold potentials

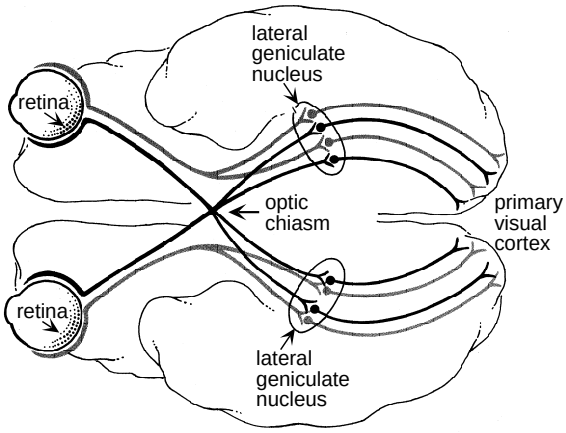

Figure 2.5 Pathway from the retina through the lateral geniculate nucleus (LGN) of the thalamus to the primary visual cortex in the human brain. (Adapted from Nicholls et al., 1992.)

of the two retinal ganglion cells shown are similar to those of the bipolar cells immediately above them in the figure, but now with superimposed action potentials. The two retinal ganglion cells shown in the figure have different responses and transmit different sequences of action potentials. G_2 fires while the light is on, and G_1 fires when it turns off. These are called ON and OFF responses, respectively. The optic nerve conducts the output spike trains of retinal ganglion cells to the lateral geniculate nucleus of the thalamus, which acts as a relay station between the retina and primary visual cortex (figure 2.5). Prior to arriving at the LGN, some retinal ganglion cell axons cross the midline at the optic chiasm. This allows the left and right sides of the visual fields from both eyes to be represented on the right and left sides of the brain, respectively (figure 2.5).

ON and OFF responses

Neurons in the retina, LGN, and primary visual cortex respond to light stimuli in restricted regions of the visual field called their receptive fields. Patterns of illumination outside the receptive field of a given neuron cannot generate a response directly, although they can significantly affect responses to stimuli within the receptive field. We do not consider such effects, although they are of considerable experimental and theoretical interest. In the monkey, cortical receptive fields range in size from around a tenth of a degree near the fovea to several degrees in the periphery. Within the receptive fields, there are regions where illumination higher than the background light intensity enhances firing, and other regions where lower illumination enhances firing. The spatial arrangement of these regions determines the selectivity of the neuron to different inputs. The term "receptive field" is often generalized to refer not only to the overall region where light affects neuronal firing, but also to the spatial and temporal structure within this region.

receptive fields

Visually responsive neurons in the retina, LGN, and primary visual cortex are divided into two classes, depending on whether or not the contribu-

simple and complex cells

tions from different locations within the visual field sum linearly. X cells in the cat retina and LGN, P cells in the monkey retina and LGN, and simple cells in primary visual cortex appear to satisfy this assumption. Other neurons, such as Y cells in the cat retina and LGN, M cells in the monkey retina and LGN, and complex cells in primary visual cortex, do not show linear summation across the spatial receptive field, and nonlinearities must be included in descriptions of their responses. We do this for complex cells later in this chapter.

A first step in studying the selectivity of any neuron is to identify the types of stimuli that evoke strong responses. Retinal ganglion cells and LGN neurons have similar selectivities and respond best to circular spots of light surrounded by darkness or dark spots surrounded by light. In primary visual cortex, many neurons respond best to elongated light or dark bars or to boundaries between light and dark regions. Gratings with alternating light and dark bands are effective and frequently used stimuli for these neurons.

Many visually responsive neurons react strongly to sudden transitions in the level of image illumination, a temporal analogue of their responsiveness to light-dark spatial boundaries. Static images are not very effective at evoking visual responses. In awake animals, images are constantly kept in motion across the retina by eye movements. In experiments in which the eyes are fixed, moving light bars and gratings, or gratings undergoing periodic light-dark reversals (called counterphase gratings) are more effective stimuli than static images. Some neurons in primary visual cortex are directionally selective; they respond more strongly to stimuli moving in one direction than in the other.

To streamline the discussion in this chapter, we consider only gray-scale images, although the methods presented can be extended to include color. We also restrict the discussion to two-dimensional visual images, ignoring how visual responses depend on viewing distance and encode depth. In discussing the response properties of retinal, LGN, and V1 neurons, we do not follow the path of the visual signal, nor the historical order of experimentation, but instead begin with primary visual cortex and then move back to the LGN and retina. In this chapter, the emphasis is on properties of individual neurons; we discuss encoding by populations of visually responsive neurons in chapter 10.

The Retinotopic Map

A striking feature of most visual areas in the brain, including primary visual cortex, is that the visual world is mapped onto the cortical surface in a topographic manner. This means that neighboring points in a visual image evoke activity in neighboring regions of visual cortex. The retinotopic map refers to the transformation from the coordinates of the visual world to the corresponding locations on the cortical surface.

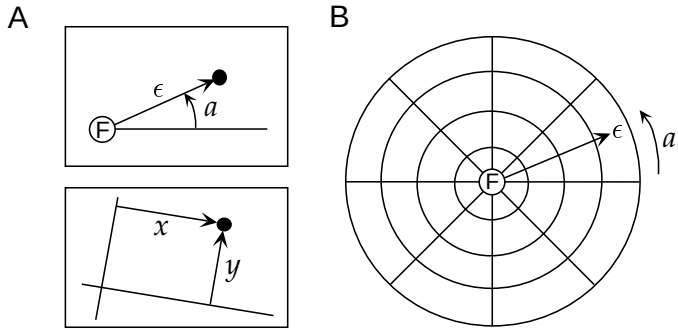

Figure 2.6 (A) Two coordinate systems used to parameterize image location. Each rectangle represents a tangent screen, and the filled circle is the location of a particular image point on the screen. The upper panel shows polar coordinates. The origin of the coordinate system is the fixation point F, the eccentricity ϵ is proportional to the radial distance from the fixation point to the image point, and a is the angle between the radial line from F to the image point and the horizontal axis. The lower panel shows Cartesian coordinates. The location of the origin for these coordinates and the orientation of the axes are arbitrary. They are usual chosen to center and align the coordinate system with respect to a particular receptive field being studied. (B) A bull's-eye pattern of radial lines of constant azimuth, and circles of constant eccentricity. The center of this pattern at zero eccentricity is the fixation point F. Such a pattern was used to generate the image in figure 2.7A.

Objects located a fixed distance from one eye lie on a sphere. Locations on this sphere can be represented using the same longitude and latitude angles used for the surface of the earth. Typically, the "north pole" for this spherical coordinate system is located at the fixation point, the image point that focuses onto the fovea or center of the retina. In this system of coordinates (figure 2.6), the latitude coordinate is called the eccentricity, ϵ, and the longitude coordinate, measured from the horizontal meridian, is called the azimuth, a. In primary visual cortex, the visual world is split in half, with the region $-90^\circ \leq a \leq +90^\circ$ for ϵ from 0° to about 70° (for both eyes) represented on the left side of the brain, and the reflection of this region about the vertical meridian represented on the right side of the brain.

eccentricity ϵ
azimuth a

In most experiments, images are displayed on a flat screen (called a tangent screen) that does not coincide exactly with the sphere discussed in the previous paragraph. However, if the screen is not too large, the difference is negligible, and the eccentricity and azimuth angles approximately coincide with polar coordinates on the screen (figure 2.6A). Ordinary Cartesian coordinates can also be used to identify points on the screen (figure 2.6A). The eccentricity ϵ and the x and y coordinates of the Cartesian system are based on measuring distances on the screen. However, it is customary to divide these measured distances by the distance from the eye to the screen and to multiply the result by $180^\circ/\pi$ so that these coordinates are ultimately expressed in units of degrees. This makes sense because it is the angular, not the absolute, size and location of an image that is typically relevant for studies of the visual system.

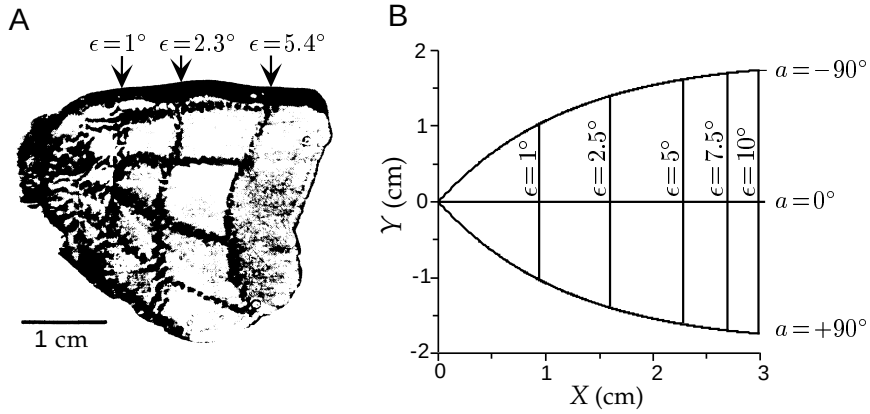

Figure 2.7 (A) An autoradiograph of the posterior region of the primary visual cortex from the left side of a macaque monkey brain. The pattern is a radioactive trace of the activity evoked by an image like that in figure 2.6B. The vertical lines correspond to circles at eccentricities of $1°$, $2.3°$, and $5.4°$, and the horizontal lines (from top to bottom) represent radial lines in the visual image at a values of $-90°$, $-45°$, $0°$, $45°$, and $90°$. Only the part of cortex corresponding to the central region of the visual field on one side is shown. (B) The mathematical map from the visual coordinates ϵ and a to the cortical coordinates X and Y described by equations 2.15 and 2.17. (A adapted from Tootell et al., 1982.)

Figure 2.7A shows a dramatic illustration of the retinotopic map in the primary visual cortex of a monkey. The pattern on the cortex seen in figure 2.7A was produced by imaging a radioactive analogue of glucose that was taken up by active neurons while a monkey viewed a visual image consisting of concentric circles and radial lines, similar to the pattern in figure 2.6B. The vertical lines correspond to the circles in the image, and the roughly horizontal lines are due to the activity evoked by the radial lines. The fovea is represented at the leftmost pole of this piece of cortex, and eccentricity increases toward the right. Azimuthal angles are positive in the lower half of the piece of cortex shown, and negative in the upper half.

cortical magnification factor

To describe the map in figure 2.7A mathematically, we write the horizontal and vertical coordinates, X and Y, describing points on the cortical surface as functions of the eccentricity ϵ and azimuth a of the corresponding points in the visual field, $X(\epsilon, a)$, $Y(\epsilon, a)$. This map is characterized by local factors, each called a cortical magnification factor, that indicate the relationship between small displacements ΔX, ΔY across the cortex and the corresponding small image displacements $\Delta\epsilon$, Δa. In general, these factors can be derived from the four elements of the Jacobian matrix, $\partial X/\partial\epsilon$, $\partial X/\partial a$, $\partial Y/\partial\epsilon$, and $\partial Y/\partial a$. However, few experiments characterize all four elements, and it is common to include additional constraints.

Figure 2.7B shows an example of such a map. Here, we assume that $X(\epsilon)$ is only a function of eccentricity, and that $Y(\epsilon, a)$ is proportional to the azimuth angle a. Further, we assume that purely radial ($\Delta a = 0$) and purely

meridional ($\Delta\epsilon = 0$) displacements have the same cortical magnification factor $M(\epsilon)$, which itself is only a function of eccentricity. Consider two nearby image points with coordinates ϵ, a and $\epsilon + \Delta\epsilon$, a, separated by an angular distance $\Delta\epsilon$. The distance separating the activity evoked by these two image points on the cortex is ΔX. By the definition of the cortical magnification factor, these two quantities satisfy $\Delta X = M(\epsilon)\Delta\epsilon$ or, taking the limit as ΔX and $\Delta\epsilon$ go to 0,

$$\frac{dX}{d\epsilon} = M(\epsilon)\,. \tag{2.13}$$

For the macaque monkey, results such as figure 2.7A suggest that

$$M(\epsilon) = \frac{\lambda}{\epsilon_0 + \epsilon}\,, \tag{2.14}$$

with $\lambda \approx 12$ mm and $\epsilon_0 \approx 1^\circ$. Integrating equation 2.13 and defining $X = 0$ to be the point representing $\epsilon = 0$, we find

$$X = \lambda \ln(1 + \epsilon/\epsilon_0)\,. \tag{2.15}$$

For purely meridional displacements, the angular distance between two points at eccentricity ϵ with an azimuthal angle difference Δa is $\Delta a \epsilon \pi/180^\circ$. Here, the factor of ϵ corrects for the increase of arc length as a function of eccentricity, and the factor of $\pi/180^\circ$ converts ϵ from degrees to radians. The separation on the cortex, ΔY, corresponding to these points has a magnitude given by the cortical amplification times this distance. Taking the limit $\Delta a \to 0$, we find that

$$\frac{dY}{da} = -\frac{\epsilon\pi}{180^\circ} M(\epsilon)\,. \tag{2.16}$$

The minus sign in this relationship appears because the visual field is inverted on the cortex. Solving equation 2.16 gives

$$Y = -\frac{\lambda \epsilon a \pi}{(\epsilon_0 + \epsilon)180^\circ}\,. \tag{2.17}$$

The map defined by equations 2.15 and 2.17 is only approximate, particularly for small eccentricities. It is also not isotropic, which means that the magnification factor $M(\epsilon)$ only describes displacements for which either $\Delta\epsilon = 0$ or $\Delta a = 0$. Nevertheless, figure 2.7B shows that these coordinates agree fairly well with the map in figure 2.7A.

For eccentricities appreciably greater than 1°, equations 2.15 and 2.17 reduce to $X \approx \lambda \ln(\epsilon/\epsilon_0)$ and $Y \approx -\lambda\pi a/180^\circ$. These two formulas can be combined by defining the complex numbers $Z = X + iY$ and $z = (\epsilon/\epsilon_0)$ $\exp(-i\pi a/180^\circ)$ (with i equal to the square root of -1) and writing $Z = \lambda \ln(z)$. For this reason, the cortical map is sometimes called a complex logarithmic map. For an image scaled radially by a factor γ, eccentricities change according to $\epsilon \to \gamma\epsilon$ while a is unaffected. Scaling of the eccentricity produces a shift $X \to X + \lambda \ln(\gamma)$ over the range of values where the simple logarithmic form of the map is valid. The logarithmic transformation thus causes images that are scaled radially outward on the retina to be represented at locations on the cortex translated in the X direction.

complex logarithmic map

Visual Stimuli

Earlier in this chapter, we used the function $s(t)$ to characterize a time-dependent stimulus. The description of visual stimuli is more complex. Gray-scale images appearing on a two-dimensional surface, such as a video monitor, can be described by giving the luminance, or light intensity, at each point on the screen. These pixel locations are parameterized by Cartesian coordinates x and y, as in the lower panel of figure 2.6A. However, pixel-by-pixel light intensities are not a useful way of parameterizing a visual image for the purposes of characterizing neuronal responses. This is because visually responsive neurons, like many sensory neurons, adapt to the overall level of screen illumination. To avoid dealing with adaptation effects, we describe the stimulus by a function $s(x, y, t)$ that is proportional to the difference between the luminance at the point (x, y) at time t and the average or background level of luminance. Often $s(x, y, t)$ is also divided by the background luminance level, making it dimensionless. The resulting quantity is called the contrast.

contrast

During recordings, visual neurons are usually stimulated by images that vary over both space and time. A commonly used stimulus, the counterphase sinusoidal grating, is described by

counterphase sinusoidal grating

$$s(x, y, t) = A\cos\big(Kx\cos\Theta + Ky\sin\Theta - \Phi\big)\cos(\omega t)\,. \tag{2.18}$$

Figure 2.8 shows a similar grating (a spatial square wave is drawn rather than a sinusoid) and illustrates the significance of the parameters K, Θ, Φ, and ω. K and ω are the spatial and temporal frequencies of the grating (these are angular frequencies), Θ is its orientation, Φ is its spatial phase, and A is its contrast amplitude. This stimulus oscillates in both space and time. At any fixed time, it oscillates in the direction perpendicular to the orientation angle Θ as a function of position, with wavelength $2\pi/K$ (figure 2.8A). At any fixed position, it oscillates in time with period $2\pi/\omega$ (figure 2.8B). For convenience, Θ is measured relative to the y axis rather than the x axis, so that a stimulus with $\Theta = 0$ varies in the x, but not in the y, direction. Φ determines the spatial location of the light and dark stripes of the grating. Changing Φ by an amount $\Delta\Phi$ shifts the grating in the direction perpendicular to its orientation by a fraction $\Delta\Phi/2\pi$ of its wavelength. The contrast amplitude A controls the maximum degree of difference between light and dark areas. Because x and y are measured in degrees, K is expressed in the rather unusual units of radians per degree and $K/2\pi$ is typically reported in units of cycles per degree. Φ has units of radians, ω is in radians/s, and $\omega/2\pi$ is in Hz.

spatial frequency K
frequency ω
orientation Θ
spatial phase Φ
amplitude A

Experiments that consider reverse correlation and spike-triggered averages use various types of random and white-noise stimuli in addition to bars and gratings. A white-noise stimulus, in this case, is one that is uncorrelated in both space and time so that

white-noise image

$$\frac{1}{T}\int_0^T dt\, s(x, y, t)s(x', y', t+\tau) = \sigma_s^2\delta(\tau)\delta(x - x')\delta(y - y')\,. \tag{2.19}$$

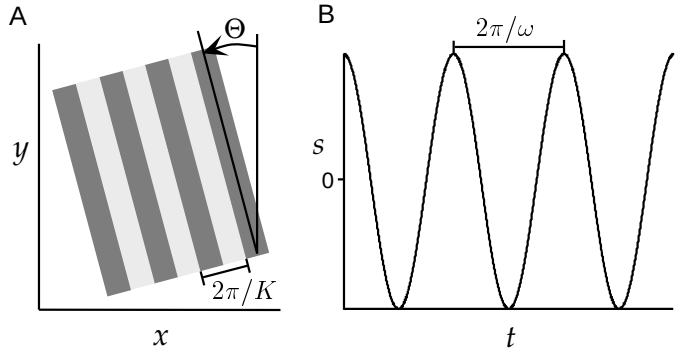

Figure 2.8 A counterphase grating. (A) A portion of a square-wave grating analogous to the sinusoidal grating of equation 2.18. The lighter stripes are regions where $s > 0$, and $s < 0$ within the darker stripes. K determines the wavelength of the grating and Θ, its orientation. Changing its spatial phase, Φ, shifts the entire light-dark pattern in the direction perpendicular to the stripes. (B) The light-dark intensity at any point of the spatial grating oscillates sinusoidally in time with period $2\pi/\omega$.

Of course, in practice a discrete approximation of such a stimulus must be used by dividing the image space into pixels and time into small bins. In addition, more structured random sets of images (randomly oriented bars, for example) are sometimes used to enhance the responses obtained during stimulation.

The Nyquist Frequency

Many factors limit the maximal spatial frequency that can be resolved by the visual system, but one interesting effect arises from the size and spacing of individual photoreceptors on the retina. The region of the retina with the highest resolution is the fovea at the center of the visual field. Within the macaque or human fovea, cone photoreceptors are densely packed in a regular array. Along any direction in the visual field, a regular array of tightly packed photoreceptors of size Δx samples points at locations $m\Delta x$ for $m = 1, 2, \ldots$. The (angular) frequency that defines the resolution of such an array is called the Nyquist frequency and is given by

Nyquist frequency

$$K_{\text{nyq}} = \frac{\pi}{\Delta x}\,. \tag{2.20}$$

To understand the significance of the Nyquist frequency, consider sampling two cosine gratings with spatial frequencies of K and $2K_{\text{nyq}} - K$, with $K < K_{\text{nyq}}$. These are described by $s = \cos(Kx)$ and $s = \cos((2K_{\text{nyq}} - K)x)$. At the sampled points, these functions are identical because $\cos((2K_{\text{nyq}} - K)m\Delta x) = \cos(2\pi m - Km\Delta x) = \cos(-Km\Delta x) = \cos(Km\Delta x)$ by the periodicity and evenness of the cosine function (see figure 2.9). As a result, these two gratings cannot be distinguished by examining them only at the sampled points. Any two spatial frequencies $K < K_{\text{nyq}}$ and $2K_{\text{nyq}} - K$

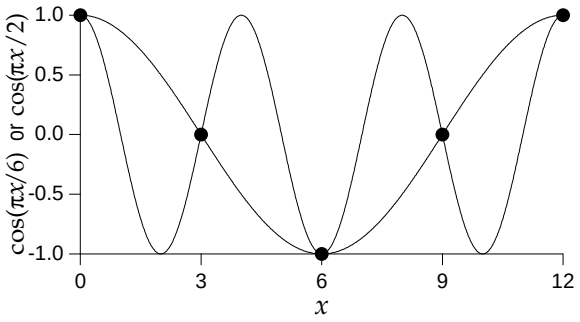

Figure 2.9 Aliasing and the Nyquist frequency. The two curves are the functions $\cos(\pi x/6)$ and $\cos(\pi x/2)$ plotted against x, and the dots show points sampled with a spacing of $\Delta x = 3$. The Nyquist frequency in this case is $\pi/3$, and the two cosine curves match at the sampled points because their spatial frequencies satisfy the relation $2\pi/3 - \pi/6 = \pi/2$.

can be confused with one another in this way, a phenomenon known as aliasing. Conversely, if an image is constructed solely of frequencies less than K_{nyq}, it can be reconstructed perfectly from the finite set of samples provided by the array. There are 120 cones per degree at the fovea of the macaque retina, which makes $K_{\text{nyq}}/(2\pi) = 1/(2\Delta x) = 60$ cycles per degree. In this result, we have divided the right side of equation 2.20, which gives K_{nyq} in units of radians per degree, by 2π to convert the answer to cycles per degree.

2.4 Reverse-Correlation Methods: Simple Cells

The spike-triggered average for visual stimuli is defined, as in chapter 1, as the average over trials of stimuli evaluated at times $t_i - \tau$, where t_i for $i = 1, 2, \ldots, n$ are the spike times. Because the light intensity of a visual image depends on location as well as time, the spike-triggered average stimulus is a function of three variables,

$$C(x, y, \tau) = \frac{1}{\langle n \rangle} \left\langle \sum_{i=1}^{n} s(x, y, t_i - \tau) \right\rangle . \tag{2.21}$$

Here, as in chapter 1, the brackets denote trial averaging, and we have used the approximation $1/n \approx 1/\langle n \rangle$. $C(x, y, \tau)$ is the average value of the visual stimulus at the point (x, y) a time τ before a spike was fired. Similarly, we can define the correlation function between the firing rate at time t and the stimulus at time $t + \tau$, for trials of duration T, as

$$Q_{\text{rs}}(x, y, \tau) = \frac{1}{T} \int_0^T dt\, \mathsf{r}(t) s(x, y, t + \tau) . \tag{2.22}$$